Review of "Instantaneous Reproduction Number Estimation From Modelled Incidence"

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This paper proposes an extension of common R_t estimation methodology to account for the fact that incidence numbers may not be directly observed, but instead are reconstructed using a statistical model. The associated uncertainty then needs to be accounted for. I find the general idea appealing and the arguments to obtain the distributions of derived quantities are elegant. However, I see some major issues with respect to the definition of the estimand and related questions of uncertainty quantification.

1. My main reservation about this paper concerns the estimand, i.e., the quantity we are aiming to estimate. A couple of points on this.

   (a) In equation (1) the authors define the instantaneous reproduction number as

   $$R_t^* = \frac{I_t}{\sum_{\tau=1}^{k} w_k I_{t-k}}. \tag{i}$$

   This is an unusual definition, and I added an asterisk to distinguish it from the standard definition. Cori et al and the related literature define R_t via the renewal equation

   $$I_t \mid I_{t-1}, I_{t-2}, \cdots \sim \text{Pois}(R_t \times \Lambda_t), \Lambda_t = \sum_{\tau=1}^{k} w_k I_{t-k}. \tag{ii}$$

   This implies

   $$R_t = \frac{\mathbb{E}(I_t \mid I_{t-1}, I_{t-2}, \dots)}{\sum_{\tau=1}^{k} w_k I_{t-k}}, \tag{iii}$$

   which looks a bit like (i). But while R_t^* is an observable quantity if the I_t are observed, R_t in (ii) it is a latent parameter that we can only estimate. The confusion likely comes from the fact that R_t^* is the maximum likelihood estimator of R_t under model (ii). But the distinction between estimand and estimator is important, and I would argue that the estimand has to be R_t as in equation (ii).

   (b) I agree that if the I_t are not directly observable we may be interested in translating our uncertainty about the I_t to the uncertainty about R_t^*. But this is only part of the way if we are really interested in R_t. In that case, we need to also take into account that even for known I_t and thus R_t^* there is estimation uncertainty about R_t. Without having figured out the details I am wondering if Rubin's rule could be used to this end (see e.g., `https://link.springer.com/article/10.1186/1471-2288-9-57`).

   (c) I am unsure about what is the point for the approximation ("≈") in equation (2), and I am wondering why it would hold. Depending on how dispersed I_t is given λ_t, this may or may not be a good approximation?

(d) I was confused by the sentence "For each parametrisation of R_t we generate 50 simulations with different random seeds, so that in bulk the simulation reproduction number will be close to the parametrised R_t ." If the distinction between estimand and estimator is taken seriously then this kind of argument will no longer be needed as it will just be clear what is the true value of R_t we would like to recover.

(e) All of the above are especially important if the procedure shall be compared to EpiEstim (as is currently done). Such comparisons only make sense if both methods aim to estimate the same thing.

(f) Also concerning the comparison to EpiEstim: considering uncertainties in I_t and w_t, but not the statistical estimation uncertainty is the exact opposite of what happens in EpiEstim. This makes comparisons of the uncertainty intervals quite tricky.

2. The assumption that incidences can be modelled as log-normally distributed is only appropriate in high-count settings. The Cori method, however, is designed for low-count settings (otherwise there would be no reason to use a Poisson distribution in (ii)). This limitation should be discussed.

3. A known limitation of the Cori method is that it cannot handle overdispersion (as it uses a Poisson rather than e.g., negative binomial distribution). This aspect should be discussed and overdispersion should be taken into account if the authors extend their method to account for the estimation uncertainty given the I_t.

4. I am not sure I agree with the discussion about negative serial intervals on page 4 (bottom). Maybe the formulas allow plugging in negative values, but I think this is at best a rough fix. To account for this properly one would need to integrate a deconvolution step. I would argue that in the authors' framework, this step would be part of the model that returns the (log-normally distributed) modelled incidences. So I would not discuss allowing for negative times between events at this point.

5. I found the simulation setup very hard to follow (see also comment 1c above) and was a little overwhelmed by the rapid introduction of additional statistical procedures ("Firstly we used a simple statistical Poisson model with time varying rate parameter, represented by a piecewise polynomial of order 2, and fitted using maximum likelihood with a logarithmic link function according to the methods of Loader et al. [25] and a bandwidth equivalent to 14 day [...]"). The methods assessed and the simulation setup need to be documented in more detail. In the current version I find it difficult to make specific comments on this part.

6. A somewhat related approach is the bootstrapping step used by Scire and colleagues here: `https://link.springer.com/article/10.1186/s12859-023-05428-4`. The authors may want to consult this reference and see how it relates to their work.

7. The link `https://ai4ci.github.io/ggoutbreak-paper/` does not seem to work. So currently the replication materials cannot be accessed.